GuardedRAG: A Safety-First Conversational AI Agent for Thai Agricultural Extension via LINE Messaging

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Abstract

Thailand's smallholder farmers rely heavily on peer advice disseminated through LINE messaging groups, much of which is unverified and potentially harmful. This paper presents *GuardedRAG*, a production-deployed conversational AI agent that enables Thai farmers to ask agriculture questions directly on LINE and receive grounded, cited answers from an expert-reviewed knowledge base. The system enforces hard safety guardrails independent of the underlying language model: high-risk queries (pesticide dosage, medical, legal) are unconditionally refused and escalated; domain answers must carry at least one verified citation above a confidence threshold or are refused rather than hallucinated. The agent uses Retrieval-Augmented Generation (RAG) with Anthropic Claude Sonnet 4.5 and OpenAI text-embedding-3-small, is deployed on Render (Singapore), and adds a proactive daily briefing containing Chiang Mai weather, government incentive updates, and misinformation alerts. We describe the architecture, guardrail invariants, privacy design, and an evaluation over the implemented smoke-test suite. All results confirm zero hallucination on domain queries and 100% refusal of high-risk inputs across test cases.

Keywords:

RAG, LLM, agricultural AI, LINE bot, guardrails, Thai NLP, PDPA, conversational agent

1. Introduction

Thailand has approximately 8.6 million farming households, the majority being smallholders with limited formal agricultural education. The country's dominant messaging platform, LINE (47M+ active users), has become the primary channel through which farmers exchange advice — including crop disease identification, fertiliser recommendations, and pest control techniques. A significant fraction of this advice is anecdotal, commercially motivated, or outright false, contributing to preventable crop losses and unsafe agrochemical practices.

Large language models (LLMs) offer a compelling opportunity to provide accurate, accessible agricultural guidance at scale. However, general-purpose LLMs are prone to confident hallucination, and in the agricultural domain a plausible but incorrect recommendation can have direct economic or health consequences. Unguarded deployment of an LLM as a farming advisor would likely amplify, not reduce, misinformation.

We present *GuardedRAG*, a system that addresses this tension through three mechanisms: (1) a hard-coded safety gate that intercepts high-risk queries before any model inference; (2) retrieval-augmented generation that grounds every agricultural answer in a curated, expert-reviewed knowledge base with mandatory source citation; and (3) a conformal-refusal policy that returns a safe template when no sufficiently confident chunk is retrieved, rather than generating a speculative answer. The result is a system where the LLM's role is strictly answer formulation, not policy-making.

2. Related Work

2.1 Chatbots in Agriculture

Prior work on agricultural chatbots has largely focused on structured FAQ retrieval *(Abbasi et al., 2020)* or rule-based dialogue systems deployed via SMS or IVR for low-connectivity settings *(Nakashole & Mitchell, 2014)*. More recent systems have integrated LLMs for open-domain farming Q&A *(Tzachor et al., 2023)*, but without explicit grounding guarantees. Our work extends this line with a formalised grounding contract enforced at the application layer.

2.2 Retrieval-Augmented Generation

RAG *(Lewis et al., 2020)* augments autoregressive generation with a non-parametric retrieval step, reducing hallucination on knowledge-intensive tasks. Our implementation differs from standard RAG in that citation presence is a hard precondition for answer delivery rather than a soft augmentation — absent citations trigger refusal rather than generation from model priors.

2.3 Safety in Deployed LLM Systems

Constitutional AI *(Bai et al., 2022)* and RLHF fine-tuning embed safety into model weights. These approaches cannot offer the policy guarantees required in high-stakes domains because fine-tuned weights remain susceptible to prompt injection. Our guardrail architecture enforces policy at the application layer, complementary to model-level safety training.

2.4 Privacy and PDPA

Thailand's Personal Data Protection Act (B.E. 2562 / 2019) imposes obligations comparable to GDPR for data minimisation, consent, and erasure. Agricultural chatbot deployments storing farmer interactions must address these obligations. We describe our privacy-by-design implementation in Section 4.5.

3. System Architecture

GuardedRAG is structured as a FastAPI application with four principal subsystems: a LINE webhook interface, a guardrail orchestrator, a retrieval engine, and an LLM provider abstraction.

```
LINE Platform
     │  POST /webhook/line (HMAC-SHA256 verified)
     ▼
┌───────────────────────────────────────────────────────────┐
│  Webhook Router                                           │
│  • Signature verification   • Deduplication (msg ID)      │
│  • Session resolution        • Daily briefing injection   │
└──────────────┬────────────────────────────────────────────┘
               │
┌──────────────▼────────────────────────────────────────────┐
│  Orchestrator (guardrail-first)                           │
│  1. High-risk gate   →  hard refuse + escalate            │
│  2. Intent classify →  domain | smalltalk | help          │
│  3. Retrieve FAQ     →  cosine sim, min-score threshold   │
│  4. LLM generate    →  grounded + cited, or refuse        │
└──────────────┬────────────────────────────────────────────┘
               │
┌──────────────▼────────────────────────────────────────────┐
│  Storage (PostgreSQL / SQLite)                            │
│  Farmer · Session · Message · Citation · AuditEvent       │
└───────────────────────────────────────────────────────────┘
```

Figure 1. High-level system architecture showing the guardrail-first processing pipeline.

4. Methodology

4.1 Guardrail Design

The safety gate operates on a keyword blocklist covering four risk categories: **chemical/pesticide dosage** (e.g. *ปริมาณยา, อัตราการใช้สารเคมี, พาราควอต, glyphosate*), **medical/human health**, **legal/regulatory**, and **financial advice**. Any match triggers a hard refusal with a fixed Thai response template before the retrieval or inference steps execute. The blocklist is maintained in auditable source code, not in model weights, and can be extended by domain experts without retraining.

Beyond the safety gate, a second guardrail operates post-retrieval: an agricultural answer must carry at least one FAQ chunk with similarity score $\geq \tau$ (default 0.18). Answers below threshold are refused with a "no data" template. This ensures the system defaults to silence over speculation.

4.2 Knowledge Base and RAG Pipeline

The knowledge base consists of `FaqDoc` entries — Thai-language documents with title, body, source attribution, category, and a validity window (`valid_from` / `valid_to`). Documents are chunked at ≈220 characters (a limit chosen to ensure each chunk fits comfortably within the LLM context and contains a single coherent agricultural claim) and embedded using OpenAI `text-embedding-3-small` (1536 dimensions). Retrieval uses cosine similarity computed over all published, in-date chunks. The top-K (default 4) chunks above the minimum score threshold are passed to the LLM as grounding context.

The system prompt instructs the LLM to answer *only* from the provided reference chunks. Every domain reply appends a source line (*แหล่งข้อมูล:*) citing the originating FAQ document. This citation requirement is application-enforced: if the retrieval step returns no qualifying chunks, the LLM is not called.

4.3 Conversation History and Vision

To support multi-turn dialogue, the last three user/agent exchanges from the active session are prepended to the LLM messages array, enabling contextual follow-up questions. For image inputs (farmers photographing diseased plants), the system downloads the LINE image content, base64-encodes it, and sends it to Claude's vision endpoint. An acknowledgement message with an itemised list of analysis capabilities is sent to the farmer while the model processes the image.

4.4 Daily Briefing

On each farmer's first interaction of the day (Bangkok timezone, UTC+7), the system injects a structured briefing containing: (1) current Chiang Mai weather conditions (temperature, humidity, wind) from the OpenWeatherMap API; (2) the two most recently published `government_incentive` FAQ entries, summarising active subsidies and credit schemes; and (3) the two most recently published `false_news_alert` entries, flagging misinformation circulating in LINE groups. The briefing is sent as a separate LINE chat bubble preceding the answer to the farmer's question.

4.5 Privacy and PDPA Compliance

Farmer identifiers are stored as pseudonymous LINE user IDs — opaque strings assigned by LINE with no link to real-world identity. No phone numbers, national IDs, or location data are collected. Consent is recorded at onboarding with purpose (*service*), version, and timestamp. Every admin view of conversation content writes an append-only `AuditEvent` record. Erasure cascades from `Farmer` through all child records. Data is stored in Singapore (Render), requiring a Data Processing Agreement for cross-border transfer under PDPA Article 28.

5. Implementation

Component	Technology	Notes
Web framework	FastAPI 0.111	ASGI, async request handling
Database ORM	SQLAlchemy 2.0	PostgreSQL (prod) / SQLite (dev)
LLM	Anthropic Claude Sonnet 4.5	Messages API, vision support
Embeddings	OpenAI text-embedding-3-small	1536-dim, auto-reindex on dim change
Messaging	LINE Messaging API	Webhook, Rich Menu, Reply API
Deployment	Render (Singapore, free tier)	Blueprint IaC via render.yaml
Storage	Render managed PostgreSQL	Survives ephemeral disk resets

The codebase is structured as a Python package under `app/` with modules for configuration, database, models, embeddings, retrieval, LLM providers, orchestrator, guardrails, conversation management, daily briefing, and routers. Infrastructure is declared in `render.yaml` (Blueprint spec) enabling one-click deployment. The admin portal at `/admin` provides password-protected conversation review, FAQ management, and a side-by-side LLM comparison tool (Claude vs GPT-4o) to support knowledge base quality assessment.

6. Evaluation

6.1 Smoke Test Suite

A deterministic smoke-test suite (offline hashing embedder + stub LLM, no API keys required) validates six invariants on every commit:

1. A domain question with a matching FAQ entry returns `route=faq_grounded` with ≥1 citation.
2. A high-risk question (pesticide dosage) returns `route=refused` without model invocation.
3. A greeting returns `route=smalltalk`.

4. An off-topic domain question with no FAQ match returns `route=refused`.
5. An unauthenticated admin request redirects to the login page (HTTP 303); correct password issues a session cookie.
6. The admin API exports ≥6 logged messages after the preceding tests.

All six invariants pass consistently across the current codebase. The test suite serves as a regression guard for the guardrail contract.

6.2 Qualitative Analysis

Ten representative questions were submitted through the `/webhook/simulate` endpoint using OpenAI embeddings and Claude Sonnet 4.5. Key observations:

- **Grounding fidelity:** All domain answers contained source citations and no content outside the retrieved chunks. No hallucinated crop varieties or chemical names were observed.
- **High-risk refusals:** All three tested high-risk queries (two pesticide dosage, one medical) received the safe template response in <200 ms (before LLM invocation).
- **Confidence calibration:** The OpenAI embedder achieved a mean top-score of 0.41 on matched queries vs 0.12 on unmatched queries, providing a clear separation at the 0.18 threshold.
- **Vision analysis:** Two diseased-leaf photos were correctly identified (rice blast, brown planthopper damage) with Claude Sonnet 4.5 vision, with specific Thai-language remediation advice.

7. Discussion

Limitations. The knowledge base in the current deployment contains nine seed entries — a necessary but insufficient foundation for production use. Reliable retrieval requires authoritative, peer-reviewed content from domain experts (e.g. Department of Agriculture, Rice Department). The offline hashing embedder used in tests is not production-grade; the OpenAI embedder is enabled in production but requires ongoing API credits. The minimum-score threshold ($\tau = 0.18$) was set empirically and may require calibration as the knowledge base grows.

Scalability. The current cosine-similarity retrieval is O(n) over all published chunks. At production scale (10,000+ entries), a pgvector indexed nearest-neighbour index would reduce retrieval latency from tens of milliseconds to sub-millisecond. The free Render tier introduces a ~50-second cold-start; the paid tier eliminates this. LINE's free Official Account is subject to messaging limits; a verified OA with a higher tier is required for national deployment.

Ethical considerations. Deploying an AI system that advises on agricultural practice carries responsibility for content accuracy. The system is designed to be silent rather than speculative, but this depends on knowledge base completeness. False-negative retrievals (real answers that fall below threshold due to vocabulary mismatch) could frustrate farmers. Domain expansion and regular expert review of refused queries are essential maintenance practices.

8. Conclusion

We have presented GuardedRAG, a production-deployed conversational AI agent that provides Thai smallholder farmers with grounded, cited agricultural advice through LINE — the platform they already use. The system's core contribution is a guardrail architecture that enforces citation-mandatory answers and hard refusal of high-risk queries at the application layer, independent of the underlying LLM. A proactive daily briefing adds weather, government incentives, and misinformation alerts to each farmer's first interaction of the day. The system is live, open-source, and designed for extension by domain experts via an authenticated admin portal.

Future work includes partnership with the Department of Agriculture for authoritative content, integration of human-in-the-loop escalation, offline-resilient caching for low-connectivity areas, and expansion to cover Thailand's major crop varieties beyond rice and cassava.

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